UE24CS352A Machine Learning – Mini-Project

Crowdfunding: Predicting Kickstarter Project Success

NAME: Raksha R Poojary

SRN: PES2UG24AM214

Name : Mithun S

SRN : PES2UG24AM208

1. Problem Statement:

The objective of this project is to predict whether a Kickstarter crowdfunding project will be successful or not successful using

information available around the project's launch. This is formulated as a binary classification problem. The model uses funding

goal, campaign duration, category, country, currency, launch-date information, project name, and project description. The aim is to

identify patterns in historical Kickstarter projects that can help estimate the likelihood of success for a new project.

2. Dataset:

The project uses Kickstarter datasets collected from Web Robots. Five 2026 snapshots were combined: 2026-05-12, 2026-06-11,

2026-07-24, 2026-08-12, and 2026-09-10. The combined data contained 351 CSV files, 1,093,953 raw rows, and 42 columns. The

same project can occur in multiple snapshots or states, so duplicate handling and project-level outcome selection were performed

before model training.

Dataset measure	Value

Raw CSV files	351
Raw rows	1,093,953
Final projects	204,398
Successful projects	123,064 (60.21%)
Not-successful projects	81,334 (39.79%)

3. Approach:

Data cleaning. All CSV files are combined and exact duplicate rows are removed. Duplicate project-state records are handled

using the project ID and state. A project is assigned target 1 if a successful outcome is present. If there is no successful outcome

but a completed negative outcome (failed, canceled, or suspended) is present, the target is 0. Projects with only live, started, or

submitted states are excluded because their final outcome is not available. One record is retained per project for the final dataset.

The goal is converted to USD and campaign duration and launch-date features are derived.

Features. Five numerical features are used: goal in USD, campaign duration, launch year, launch month, and launch day of week.

Three categorical features are used: category, country, and currency. The project name and blurb are combined into a text feature.

TF-IDF with unigram and bigram features is used for text; one-hot encoding is used for categorical variables; numerical variables

are standardized.

Models. Two classification models are trained: Logistic Regression and Random Forest. The dataset is divided using an 80/20

stratified train-test split with random_state = 42. This produces 163,518 training projects and 40,880 test projects. The

preprocessing steps are included inside the model pipelines so the same transformations are applied during training and prediction.

4. Implementation Overview

File	Purpose
data_cleaning.py	Combines raw CSVs, removes duplicates, determines outcomes, creates the cleaned dataset.
preprocessing.py	Creates numerical, categorical and TF-IDF text preprocessing.
train_model.py	Splits the data and trains Logistic Regression and Random Forest.
evaluate_model.py	Evaluates both trained models on the held-out test set and saves metrics/graph.
predict.py	Accepts a new project's details and predicts success using the trained Random Forest.

5. Results:

The models were evaluated on 40,880 projects that were not used to fit the models. Accuracy, precision, recall, and F1-score were
calculated. The results show that Random Forest performs better than Logistic Regression on the held-out test set.

Model	Accuracy	F1 (Success)	Precision (Success)	Recall (Success)
Logistic Regression	79.94%	83.61%	82.27%	85.00%
Random Forest	81.29%	84.85%	82.81%	86.99%

6. Prediction Demonstration:

The final prediction program accepts the details of a new project, including its name, description, funding goal, campaign duration,
launch date, category, country, and currency. A hypothetical Smart Water Bottle project was entered as a new input and the
Random Forest produced a prediction of SUCCESSFUL with an estimated probability of 97.0%. This 97.0% is the model's
estimated probability for that individual input; it is not the overall model accuracy. Overall performance is measured using the
held-out test set, where Random Forest achieved 81.29% accuracy.

7. Conclusion:

The project implements a complete machine learning workflow for Kickstarter success prediction, beginning with raw-data cleaning

and ending with prediction on a new project. The final dataset contains 204,398 projects after duplicate handling and outcome

filtering. Logistic Regression provides a baseline with 79.94% test accuracy, while Random Forest achieves 81.29% and is

therefore selected for the prediction demonstration. The results show that launch-time metadata and project text can be combined

to build a useful classification model.

8. Future Scope:

Future improvements could include hyperparameter tuning, additional time-aware validation, richer natural-language features, and

more advanced text models. These are possible extensions and are not part of the current implementation.